

Does Intermunicipal Cooperation Decrease the Property Tax Gap?

Evidence from Italy

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Local governments face a fundamental trade-off: political representation favours small jurisdictions; economies of scale in complex functions favour large ones.



Some small municipalities resolve it by joining forces — pooling staff and expertise without giving up political independence. That is **Intermunicipal Cooperation (IMC)**.



We ask: does IMC help municipalities reduce tax evasion and improve collection of what they are owed?



Short answer: **yes**, but only where enforcement capacity is structurally insufficient.

Motivation & Contribution

The IMC literature focuses on expenditure:

- Cost savings in waste, transport, public services — mixed results (Bel, 2015, 2016)
- Quality improvements emerging (Gori et al., 2025)

The revenue side is (almost) neglected:

- Small municipalities have **statutory** taxing power ... but often lack **enforcement capacity**
- Notsu et al. (2025) — IMC improves tax debt recovery in Japan. No *causal* evidence on evasion; no Italian evidence.

This paper

Quasi-experimental evidence that IMC affects **revenue enforcement** — using novel official data on Italy's property tax gap.

The Italian Property Tax: IMU

IMU is the main local tax on real estate, and the most important source of local revenue.

What is taxed: second homes, business premises, rented dwellings, luxury primary residences, non-exempt agricultural land
(primary residence **exempt**).

Rate structure: Standard rate 8.6‰ set nationally; municipalities adjust within predefined bands $\Rightarrow$ statutory **autonomy**

The enforcement problem

The cadastral register determines the taxable base — but municipalities **lack the personnel to verify which properties are taxable and to audit non-compliant owners**.

Result: **statutory power $\neq$ effective collection**

The Tax Gap: Two Components, One Problem

$$\text{Tax Gap} = \underbrace{(\overset{\text{unobservable}}{\downarrow} \text{Theoretical} - \text{Assessed})}_{\text{Assessment Gap (Evasion)}} + \underbrace{(\text{Assessed} - \text{Collected})}_{\text{Collection Gap (Residuals)}}$$

Outcome variable — Ministry of Economy and Finance (MEF) official estimate:

$$\text{IMU Tax Gap}_i = 1 - \frac{\text{Standardized Revenue}}{\text{Cadastral Value} \times \text{Standard Rate}}$$

Both components reflect the municipality's **inability to enforce its statutory rights**. MEF computes this via cadastral cross-checks unavailable to small municipalities.

IMC in Italy: The Legal Framework (TUEL 267/2000)

IMC was designed to nudge small municipalities (<5,000 inhabitants) to cooperate without giving up political independence — reducing the political conflict the central government faces when forcing mergers.

Convenzione (Art. 30):

- Contractual agreement
- No new legal entity
- Function-specific, renewable

Unione di Comuni (Art. 32):

- Autonomous public body
- Own budget and statutory organs
- Permanent joint management

Key advantage over mergers: pools enforcement capacity while retaining **political autonomy and separate electoral accountability**

Regulatory timeline: 2000: voluntary → 2010: mandatory (<5,000 inhab.) → 2024: voluntary again (incentives retained)

Treatment Definition

IMC can jointly manage any municipal function: Registry, Nurseries, Education, Police, Waste Management, Social Services, Public Land, Transportation, Traffic Management, Taxation, Technical Office

Treated = municipality in an IMC covering **Registry + Taxation + Technical Office**

enforcement chain: cadastral records → tax assessment → technical verification

Not-Treated = municipality not in any IMC (pure non-cooperator)

Data Sources

- **SOSE** — Municipal organizational structure, 2013–2022 (missing: 2014, 2020)
- **MEF** — Official IMU tax gap estimates, 2017–2022 (missing: 2020)
- **ISTAT** — Socio-demographic and fiscal controls

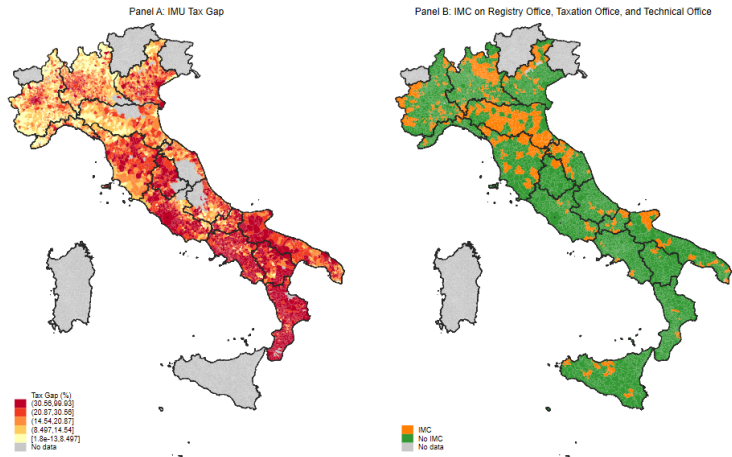
Panel: 5,062 municipalities $\times$ 5 outcome years (2017, 2018, 2019, 2021, 2022)

Descriptive Statistics: Treated vs. Not-Treated (2019)

	Not-Treated		Treated	
	Mean	SD	Mean	SD
IMU Tax Gap	0.222	0.143	0.147	0.106
North (=1)	0.534	0.499	0.753	0.432
Urbanisation (1=rural, 3=urban)	2.530	0.594	2.681	0.478
Population	9,900	57,830	4,402	7,467
Population density (per km ²)	399.3	806.8	199.8	332.1
Avg. total income (EUR)	17,714	3,840	18,374	3,187
Avg. employee income (EUR)	19,314	3,991	20,164	3,184
Schools per 1,000 inhab.	1.258	1.023	1.249	1.178
Social expenditure p.c. (EUR)	72.3	49.7	80.2	46.3
Mayor's age	49.4	7.1	49.7	7.3
Observations	3,948		1,114	

Source: SOSE, MEF. 2019 cross-section.

The Selection Problem: Geographic Distribution



Left: Tax gap (darker = higher). *Right:* IMC (orange) vs. non-IMC (green). Year 2019.

Empirical Strategy: Motivation

Data problem: IMC started in 2000; we have data for the outcome only from 2017. The biggest wave of adoption happened between 2013 and 2016. We do not observe the outcome before treatment for the majority of IMC municipalities.

Selection problem: municipalities self-select into IMC based on capacity, geography, and institutions $\Rightarrow$ naive OLS is **biased**.

Solution: Matching with IPW

We identify the ATT via **selection-on-observables** in a cross-sectional setup (year-by-year): we reweight the control group to match the treated covariate distribution.

Our main results are computed on the **2019** cross-section, but we check stability across all years.

Step 1 — Propensity Score via LASSO

We estimate the **propensity score** $p(X_i) = P(\text{Treated}_i = 1 \mid X_i)$ via LASSO logistic regression using pre-treatment covariates (2013). We exclude municipalities that adopted IMC before 2013 (*always treated*): their pre-treatment baseline is unobservable.

Why LASSO? Standard selection is researcher-driven and risks arbitrary choices. LASSO is data-driven: a cross-validated penalty λ shrinks irrelevant predictors to zero.

Candidate pool: 38 variables **Retained:** 21 variables

Institutional: MAQI pillars I–III (bureaucracy, politicians, fiscal performance)

Geographic: urbanization, altitude zone, coastal, island

Demographic: pop. size classes, log pop., density

Fiscal/income: tax burden, income surtax, avg. income, employee income

Regional FE **forced in** Restricted to **post-2013 cohort**

Common support result

Min-Max trimming of the propensity score distribution $\Rightarrow$ final sample $N = 4,149$ (see slide 23).

Step 2 — IPW-WLS Estimation

$$IMU\ Tax\ Gap_i = \beta_0 + \beta_1 North_i + \underbrace{\beta_2 (North \times Treated)_i}_{ATT, North} + \underbrace{\beta_3 (CS \times Treated)_i}_{ATT, Center-South} + \gamma' X_i + \varepsilon_i$$

- $North_i$: controls for baseline regional differences in the gap
- $North \times Treated$: ATT under IPW in the North
- $CS \times Treated$: ATT under IPW in the Center-South
- $\gamma' X_i$: municipality-level controls (population, density, avg. income, employee income, schools p.c., social expenditure p.c., mayor's age)

Why regional interactions? North–South divide in both gap levels and IMC take-up: a pooled coefficient would mask the key regional heterogeneity.

The IPW solution

Treated: weight = 1 (represent themselves) **Controls:** weight $\propto \hat{p}/(1 - \hat{p})$

This reweights the control group to match the treated covariate distribution.

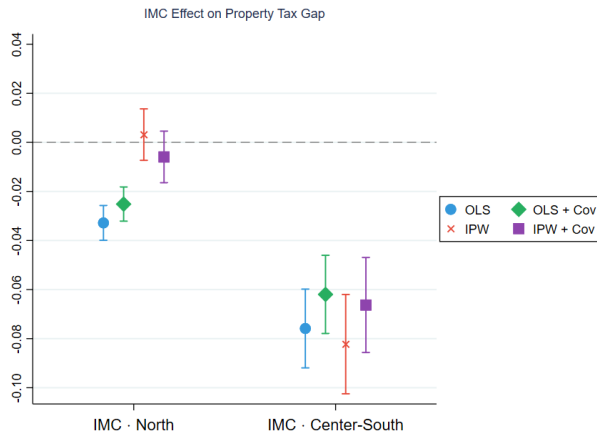
Main Results

	<i>Outcome: IMU Tax Gap</i>			
	(2) OLS	(3) OLS+Cov	(4) IPW	(5) IPW+Cov
Treated $\times$ North	−0.033*** (0.004)	−0.025*** (0.004)	0.004 (0.005)	−0.005 (0.005)
Treated $\times$ Center-South	−0.076*** (0.008)	−0.062*** (0.008)	−0.080*** (0.010)	−0.063*** (0.010)
<i>N</i>	5,062	5,062	4,149	4,149
<i>R</i> ²	.288	.356	.334	.402
SEs clustered at municipality level. *** $p < 0.01$. (year=2019).				

North: OLS shows -2.5 pp $\rightarrow$ IPW ≈ 0 . Selection, not effect. Under IPW, the North shows no average enforcement gain from IMC.

Center-South: -6.3 pp*** survives full IPW. IMC acts as a **catch-up mechanism** where enforcement capacity is structurally low.

Coefficient Plot



Results are stable across all years (2017–2022), see appendix 25.

Mechanism: Joint Taxation Office (*Ufficio Tributi*)

Test: restrict treatment to municipalities managing the *Taxation Office* through IMC specifically.

Same IPW-WLS model; treatment redefined as Tax Office only. Control pool: pure non-cooperators (municipalities with any other cooperative service excluded).

	<i>Outcome: IMU Tax Gap</i>	
	(4) IPW	(5) IPW+Cov
Tax Office $\times$ North	0.017 (0.011)	0.014 (0.010)
Tax Office $\times$ CS	-0.058*** (0.020)	-0.059*** (0.019)
<i>N</i>	3,621	3,621

244 trib-treated units; SEs clustered.

- North: **zero** n.s. — consistent with baseline
- CS: **-5.9 pp***** — confirmed channel
- Effect survives stricter treatment definition

Robustness: Spatial First Difference (SFD)

Why? IPW controls for observables. Spatially persistent unobservables — geography, institutions, local culture — may still confound.

Klein & Tchuente (2023, *Econometrics Journal*); Duranton, Gobillon & Overman (2011, *Economic Journal*)

The idea: difference each municipality against the inverse-distance-weighted (IDW) average of its $k = 4$ nearest *untreated* neighbours:

$$\Delta \text{gap}_i = \text{gap}_i - \bar{\text{gap}}_i^{\text{IDW}}, \quad w_{ij} \propto 1/d_{ij}$$

Removes unobservables shared with the nearest untreated neighbours within a $\sim 20\text{--}50$ km radius.

Estimand: local neighbourhood effect — expected to be *smaller* than the IPW ATT (different geographic scale).

Two panels:

- **Panel 1:** full untreated pool — baseline treatment
- **Panel 2:** clean control pool (pure non-cooperators) — *Taxation Office* treatment

Robustness: SFD Results

	Panel 1: Baseline Treatment		Panel 2: <i>Taxation Office</i>	
Δ Tax Gap	(1) Interact	(2) +Cov	(3) Interact	(4) +Cov
Treated $\times$ North	−0.014*** (0.003)	−0.013*** (0.003)		
Treated $\times$ CS	−0.014* (0.008)	−0.012 (0.008)		
Tax Office $\times$ North			−0.014*** (0.005)	−0.012** (0.005)
Tax Office $\times$ CS			−0.035** (0.015)	−0.032** (0.015)
<i>N</i>	5,061	5,061	4,300	4,300
Covariates		✓		✓
Control pool	Full	Full	Clean	Clean

SEs clustered at municipality level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Source: SOSE, MEF (2019).

Counterfactual Policy Simulation

If all untreated CS municipalities adopted IMC ($\approx 1,500$ potential candidates):

$$\text{Savings}_{i,t} = (-\hat{\beta}_{CS,t}) \times \text{Theoretical Tax Base}_{i,t}$$

Year	Savings (EUR M)	% of theor. rev.
2017	271.5	5.3
2018	234.7	5.0
2019	280.5	5.9
2021	325.8	6.9
2022	350.2	7.4
Total	1,462.7	

2020 excluded (COVID anomaly).

- Range: **235 – 350 M EUR/year** | 5-year total: **≈ 1.46 billion EUR**
- Savings grow as joint offices mature over time

Conclusion

1. IMC reduces the IMU gap by **−6.3 pp** in the Center-South; **zero** in the North
2. Effect survives IPW, mechanism test, SFD, and year-by-year checks
3. Driven specifically by the **joint Taxation Office** — not generic cooperation
4. Effect grows over time: joint offices mature, enforcement improves

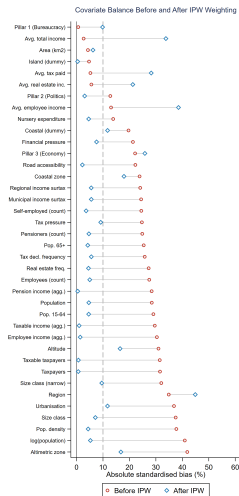
Policy: Shift from mandatory to **structurally targeted** incentives. Focus on Center-South.
Potential recovery: **235–350 M EUR/year**.

IMC is a **state-capacity-building** mechanism. Cooperation removes the enforcement constraint *where it binds most*.

Thank you. `giovanni.cuttica@unife.it`

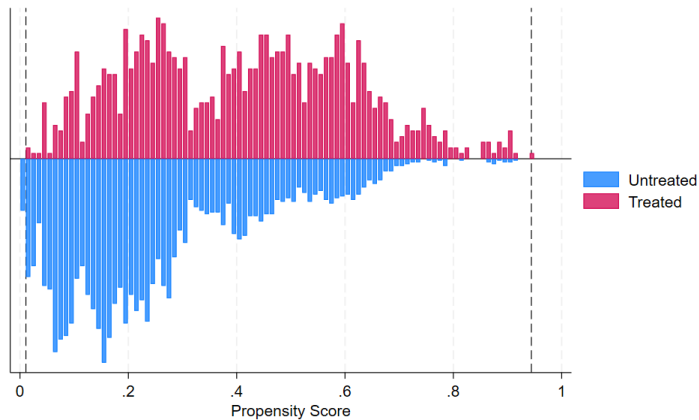
Appendix

Covariate Balance: Love Plot



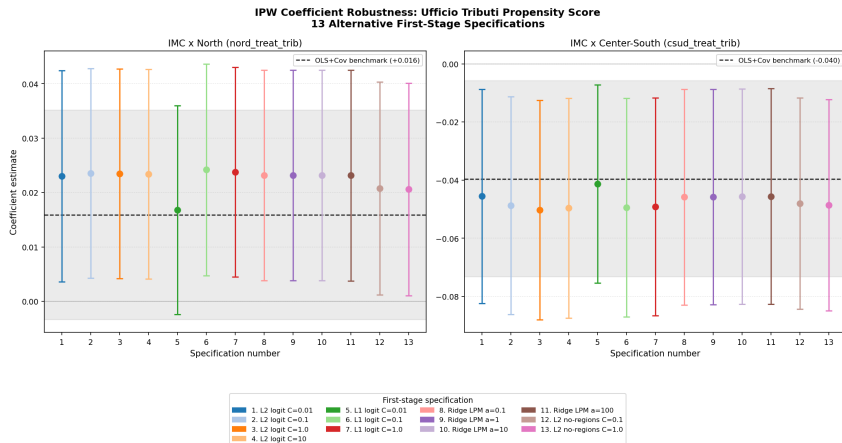
Standardized mean differences before (circles) and after (triangles) IPW reweighting for all 38 candidate covariates.

Common Support Area



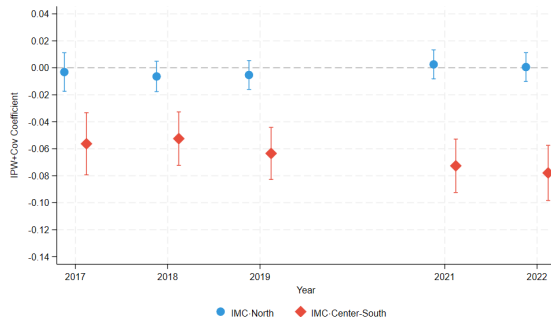
Propensity score distribution for treated (red) and control (blue). Min-Max trimming applied; final sample $N = 4,149$.

Mechanism Robustness: 13-Specification Grid



13 alternative first-stage specifications: L1/L2 logit, ridge LPM, $\pm$ regional FE. CS coefficient: -4 to -5 pp** across all 13; North: near zero n.s. throughout. Mechanism result is insensitive to propensity score estimation method.

Year-by-Year Stability



- North: **zero** in every year
- CS: **growing** — 2017: -5.6 pp → 2022: -7.8 pp
- Consistent with joint tax offices **maturing** over time
- 2019 cross-section is representative

IPW + Cov coefficients by year (2017–2022, excl. 2020).
 $N = 4,149$. 95% CIs.